

No.of Functional Features Included in the Solution

Date	14 July 2026
Team ID	SWUID2026221635
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction & Analysis)
Maximum Marks	5Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Authentication	Allows researchers and policymakers to register, login, and keep track of query histories safely.	User Account Module	Done	1.0 Mark
2	Live Parameter Inputs	Offers interactive form sliders (Life Expectancy, Schooling, GNI) for entering custom values.	Presentation Layer	Done	1.0 Mark
3	Dynamic HDI Prediction	Ingests UI input data to predict the target human development tier (Very High, High, Medium, Low).	Core Machine Learning Module	Done	1.0 Mark
4	Analytical Visualization	Generates performance gap bar charts and comparison graphs based on inputted metrics.	Visualization Engine	Done	1.0 Mark
5	Historical Log Lookup	Enables users to search, view, and comparison-track their older model predictions.	Database Storage	Done	0.5 Marks
6	Dynamic Report Export	Compiles the dashboard values and prediction charts into a downloadable PDF document.	Reporting Component	Done	0.5 Marks
7					
8					

Feature Summary:

Metric	Count / Value
Total Features Planned	6
Total Features Implemented	6
Core / Must-Have Features	4
Additional / Nice-to-Have Features	2
Features Tested&Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Web Dashboard Layout, Input Sliders	Interactive sliders to change GNI and schooling inputs.
2	Backend / Logic	Predict HDI Tier, Chart Generator	Pre-trained classifier script classifying countries dynamically.
3	Database / Storage	User Database, Prediction Logging	Mongo DB saving prediction logs and registration entries.
4	API / Integration	External API connectors	Connection framework for updating live indicators via Python Requests.
5	Security / Authentication	Login Form, Session Guard	Passwords hashed securely using the Werkzeug library.